

Bachelor in Computer Applications

Course Title: **Software Engineering**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Explain process model for software prototype development with a neat diagram focusing on its different stages.
2. Why architecture patterns are important? Explain Model-View-Controller, layered and repository architecture.
3. Explain key activities performed during Unit testing, component testing and system testing.
4. Explain the process of software configuration management.
5. Explain software review process.
6. Describe different factors affecting software pricing.
7. Draw sequence diagram for online food ordering system where the user can choose foods of various restaurants and pay online. The system also provides discount of 10% if the total amount paid is more than Rs. 5000. The user if cancels the order within 30 minutes will be approved else the paid amount will not be refunded.
8. Briefly explain requirement engineering process.
9. What are the basic principles of agile methods? Explain.
10. List various attributes of good software and explain them.
11. Describe Model-driven architecture. What type of abstract system models are produced by it?
12. Write short notes on: a) open-source development b) Test driven development